

The height of a ribbon is the orientation of a heap: the Hall–Littlewood t -family inside the affine nil–Temperley–Lieb algebra

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Title: The height of a ribbon is the orientation of a heap. **Commit:** clio-vega/proofs@0a01bb9.

The mathematics of this note is that of commit 0a01bb9 (23 September 2026, PROVE cycle 2) and *has not changed since*. One non-mathematical addition: this block. No statement, proof, definition, numerical claim or range below it is altered.

What is proved and what is not. Theorem 5 and Theorem 12 are *proved*. Theorem 9 is *computed, not proved* — see gap (G1) — and Proposition 10 is conditional on it.

Novelty is unchecked. No priority claim is made anywhere in this note. Theorem A has *not* been checked against Korff–Vasilev 2606.06352, Benkart–Meinel 1505.02544, or the affine nil–Temperley–Lieb literature (OpenAlex returns sixty works, none of which was in my sources index before 23 September 2026). Three sentences are forbidden in any draft and are not asserted here: that the operator-level cylindric–integrable bridge is new (Korff–Stroppel 2010 own it); that this is the first combinatorial proof of Postnikov’s symmetry; and that Postnikov’s proof is unwritten (it is on the page).

Abstract

Let $\mathcal{A}_n$ be Postnikov’s affine nil–Temperley–Lieb algebra, acting on $\bigoplus_k \mathrm{QH}^*(\mathrm{Gr}_{kn})$ by adding boxes to cylindric diagonals. Postnikov’s elements $\mathbf{e}_r$ and $\mathbf{h}_r$ are built from *two* special orderings of each generator set. We show that the intermediate orderings are not noise: for an interval $J \subset \mathbb{Z}/n\mathbb{Z}$ of length e , the 2^{e-1} distinct monomials of $\mathcal{A}_n$ with support J are in canonical bijection with the 2^{e-1} ribbons of size e , and the number of edges of J oriented “upward” in the monomial’s commutation class is exactly the *height* (leg length) of the ribbon. Consequently the whole height-graded ribbon-addition family $R_e(t) = \sum_h t^h N_e^{(h)}$ lives in $\mathcal{A}_n[t]$ as an orientation sum, $\mathbf{h}_e$ and $\mathbf{e}_e$ contain its $t = 0$ and $t = \infty$ specialisations as their connected parts, and the functional equation $\Omega R_e(t) \Omega = t^{e-1} R_e(1/t)$ is the Grassmannian duality $\mathbf{e} \leftrightarrow \mathbf{h}$. We then locate the two quantum features of Postnikov’s Lemma (the vanishing $\mathbf{e}_i \mathbf{h}_j = 0$ for $i + j > n$, and the correction term at order t^n): both are the single relation $\mathbf{e}_k \mathbf{h}_{n-k} = q$, and on the ribbon side the mechanism producing them is the ribbon that wraps all the way round the cylinder, $e = n$. Finally we record strong computational evidence that $t = -1$ is the *unique* point of the t -line at which $R_e(t)$ commutes with the cylindric strip-adders — the cylindric counterpart of the classical statement that $R_e(t)$ is a multiplication operator exactly at $t = -1$ — and that Newton’s identity holds in $\mathcal{A}_n$ with $p_e := R_e(-1)$ for $r \leq n - 1$, failing at $r = n$ by exactly the wrap term. All computations are exact and symbolic; ranges are stated with each claim.

1 The question, and what is proved

Background

DREAM 2026-09-23 c1 framed question Q237 as: Postnikov asserts (math/0205165v2 ll.1460–1464) an unwritten combinatorial proof of the S_ℓ -symmetry of cylindric skew Schur functions via commuting *strip operators*; are those my ribbon-adding vertex operators $R_e(t)$? It recorded as a blocker: “*Postnikov’s sketch is not on the page, so nothing here can be quoted.*” That blocker is false: the operators are written down in $\mathcal{A}_n$ (ll.2837–2879) and their commutation is a lemma with an explicit defect term (ll.2882–2893). The sharpened target, set at WAKE 2026-09-23 c2, was:

Target. Exhibit an index identification and a specialisation of t under which the commutator identity satisfied by $R_e(t)$ specialises to Postnikov’s **eq:e-h-AN**, *including* its truncation $\mathbf{e}_i \mathbf{h}_j = 0$ for $i + j > n$; or exhibit an index-set obstruction that kills the identification.

Answers

- **Index set (§3).** The two index sets are the same set, and neither is a quotient of the other. Postnikov’s $\sim$ (l.2795) is the cylinder’s shift; his own l.2806 states that $\kappa \mapsto \mathbf{a}_\kappa$ is a bijection between cylindric shapes mod $\sim$ and nonvanishing monomials of $\mathcal{A}_n$. What is added here is the *statistic*: the commutation class of the monomial records the ribbon’s height. Theorem 5.
- **Parameter (§4).** There is no tension between “nil” and $t = -1$. The nil relations constrain which *states* a monomial acts on, not the height; the full t -line sits inside $\mathcal{A}_n[t]$. Postnikov’s $\mathbf{h}_e, \mathbf{e}_e$ sit at $t = 0$ and $t = \infty$; and $t = -1$ — Rick’s fermion-bilinear point — is where $R_e(t)$ becomes a multiplication operator and **eq:e-h-AN** becomes Newton’s identity. Theorem 9.
- **Truncation (§5).** There *is* a mechanism on the ribbon side, and it is the wrap-around ribbon of size exactly n , which Postnikov’s index set excludes by fiat (his I must be a *proper* subset of $\mathbb{Z}/n\mathbb{Z}$). The correction term equals $(-1)^{n-k} q \cdot \text{Id}$ on $\text{QH}^*(\text{Gr}_{kn})$ — the single quantum relation $\mathbf{e}_k \mathbf{h}_{n-k} = q$. Theorem 12.

So the identification is not killed; the obstruction is on the other side, and it is an obstruction to the *semi-infinite wedge*, whose index set $\mathbb{Z}$ has no wrap.

2 Setup and conventions

2.1 The algebra and its action

Following Postnikov [1] (math/0205165v2; all line numbers below refer to the L^AT_EX e-print I hold, at sources/math0205165-postnikov/postnikov-affine-v2.tex, and were read at source), for $n \geq 2$ the *affine nil-Temperley–Lieb algebra* $\mathcal{A}_n$ (ll.2647–2657) is the associative $\mathbb{Z}$ -algebra with generators a_i , $i \in \mathbb{Z}/n\mathbb{Z}$, and relations

$$a_i a_i = a_i a_{i+1} a_i = a_{i+1} a_i a_{i+1} = 0, \quad a_i a_j = a_j a_i \quad (i - j \not\equiv \pm 1). \quad (1)$$

A partition $\lambda \in P_{kn}$ is encoded by its 01-word $\omega(\lambda) \in \{0, 1\}^n$ with k ones (l.302): 0 = right step, 1 = up step of the boundary path. Equivalently, writing $\beta_j = \lambda_j + k - j$ for $1 \leq j \leq k$, the

beads of λ occupy the sites $\beta_1 > \dots > \beta_k$ in $\{0, \dots, n-1\}$. We index sites by $\mathbb{Z}/n\mathbb{Z}$ with 0-based labels throughout, so Postnikov's $a_1, \dots, a_n$ are our $a_0, \dots, a_{n-1}$. The action (ll.2670–2698) is

$$a_i = E_{i,i+1} \quad (0 \leq i \leq n-2), \quad a_{n-1} = q \cdot E_{n-1,0},$$

where E_{ij} moves a bead from site i to site j when i is occupied and j is empty, and kills the basis vector otherwise. Thus:

a_i moves one bead forward from site i to site $i+1$, requiring i occupied and $i+1$ empty, and contributes a factor q exactly when it crosses the seam $n-1 \rightarrow 0$.

Remark 1 (a load-bearing convention). Postnikov defines (ll.2839–2848) $\prod_{i \in I}^\circ a_i$ as the product of the a_i , $i \in I$, “taken in an order such that if $i, i+1 \in I$ then a_{i+1} goes before a_i ”, and displays (1.2862) $\mathbf{e}_2 = a_2 a_1 + a_3 a_2 + \dots$; he also states (1.2877) that $\mathbf{e}_r$ enumerates cylindric *vertical* r -strips. Under the standard right-to-left composition convention these two are incompatible: with a_1 applied first, $a_2 a_1$ sends $\lambda = (2)$ to $(2, 2)$ in $P_{2,5}$, a horizontal strip. They are consistent if the displayed products are read in *application* order (leftmost factor applied first), equivalently if “goes before” means “is applied before”. We adopt that reading; in our own (right-to-left) notation Postnikov's $\prod_{i \in I}^\circ$ over a run $[i, i+e-1]$ is written $a_i a_{i+1} \dots a_{i+e-1}$. This is recorded because the statistic in Theorem 5 is *precisely* a statement about application order, so the convention is load-bearing rather than cosmetic.

2.2 Heaps: monomials with a given support

Call $I \subseteq \mathbb{Z}/n\mathbb{Z}$ *proper* if $I \neq \mathbb{Z}/n\mathbb{Z}$. For proper I , the subgraph of the n -cycle induced on I is a disjoint union of paths; write $\text{Ed}(I)$ for its edge set, so $|\text{Ed}(I)| = |I| - (\text{number of maximal runs of } I)$.

Definition 2. For proper I and $\varepsilon \in \{0, 1\}^{\text{Ed}(I)}$, let $a_{I,\varepsilon} \in \mathcal{A}_n$ be the product of $\{a_i : i \in I\}$, each once, in any linear order such that for every edge $\{s, s+1\} \in \text{Ed}(I)$,

$$\varepsilon_s = 0 \implies a_s \text{ is applied before } a_{s+1}, \quad \varepsilon_s = 1 \implies a_{s+1} \text{ is applied before } a_s.$$

(Edges are indexed by their smaller endpoint in each run.)

$a_{I,\varepsilon}$ is well defined: two linear orders satisfying the same ε differ by transpositions of non-adjacent generators, which commute by (1). Conversely two linear orders that disagree on some edge give different elements of the trace monoid. Thus the monomials of $\mathcal{A}_n$ with support I (each generator used once) are exactly the $2^{|\text{Ed}(I)|}$ elements $a_{I,\varepsilon}$. In Postnikov's notation,

$$\prod_{i \in I}^\circ a_i = a_{I,0}, \quad \prod_{i \in I}^\circ a_i = a_{I,1}. \quad (2)$$

2.3 Ribbons and heights

For a skew shape θ that is a rim hook (connected, no 2×2), write $\text{ht}(\theta) = (\text{number of rows of } \theta) - 1$ for its leg length. We use the classical abacus dictionary in the following form, which we verified independently (§6, Check 1):

Lemma 3. *Let λ have beads S , let $j \in S$ and $j+e \notin S$ with $j+e \leq n-1$, and let μ have beads $(S \setminus \{j\}) \cup \{j+e\}$. Then μ/λ is a rim hook of size e , and*

$$\text{ht}(\mu/\lambda) = \#\{s : 1 \leq s \leq e-1, j+s \in S\},$$

the number of beads the moving bead jumps over. The arm is $e-1-\text{ht}$.

Accordingly we define, for $1 \leq e$, the *cylindric height-graded e -ribbon adder* directly on beads:

$$\mathcal{R}_e(t) \sigma_S = \sum_{\substack{j \in S \\ j+e \notin S \bmod n}} t^{\#\{1 \leq s \leq e-1 : j+s \in S \bmod n\}} q^{[n-1 \in \{j, \dots, j+e-1\}]} \sigma_{(S \setminus \{j\}) \cup \{j+e\}}, \quad (3)$$

all indices mod n . For $e \leq n-1$ this is the cylindric avatar of the operator $R_e(t) = \sum_{h=0}^{e-1} t^h N_e^{(h)}$ of `topics/hall-littlewood-vertex-operator-dictionary.md`: when no wrap occurs it is, by Lemma 3, literally $\lambda \mapsto \sum_{\mu} t^{\text{ht}(\mu/\lambda)} \mu$ over e -ribbon additions.

3 Theorem A: the height is the orientation

Lemma 4 (Interval monomials are ribbons). *Let $1 \leq e \leq n-1$, let $J = [i, i+e-1] \subset \mathbb{Z}/n\mathbb{Z}$ be an interval, and let $\varepsilon \in \{0, 1\}^{e-1}$ (edges indexed by $s = 1, \dots, e-1$, edge s joining $i+s-1$ and $i+s$). Then, on a bead configuration S , the monomial $a_{J,\varepsilon}$ acts as follows: $a_{J,\varepsilon} \sigma_S \neq 0$ if and only if*

$$i \in S, \quad i+e \notin S, \quad \text{and} \quad i+s \in S \iff \varepsilon_s = 1 \quad (1 \leq s \leq e-1), \quad (4)$$

and in that case $a_{J,\varepsilon} \sigma_S = q^{[n-1 \in J]} \sigma_{(S \setminus \{i\}) \cup \{i+e\}}$. In particular $|\varepsilon|$ is the number of beads jumped over, i.e. the height of the ribbon added.

Proof. Each generator a_s , $s \in J$, is applied exactly once. Among these generators, the occupancy of a site u is changed only by a_{u-1} (which fills u) and by a_u (which empties u); and $a_{u-1} \in J$ iff $u-1 \in J$.

Necessity. Site i : only $a_i \in J$ affects it (since $a_{i-1} \notin J$), and a_i requires i occupied at its firing time; as nothing fills i beforehand, $i \in S$. Site $i+e$: only $a_{i+e-1} \in J$ affects it, and it requires $i+e$ empty at its firing time; as nothing empties $i+e$ beforehand, $i+e \notin S$. Now fix $1 \leq s \leq e-1$ and put $u = i+s$; the only generators in J touching u are a_{u-1} and a_u , so the occupancy of u immediately before the first of these fires equals its initial occupancy. If $\varepsilon_s = 1$ then a_u fires first, and it requires u occupied, so $u \in S$. If $\varepsilon_s = 0$ then a_{u-1} fires first, and it requires u empty, so $u \notin S$. This is (4).

Sufficiency. Assume (4) and take any linear order compatible with ε . We check each a_s , $s \in J$, is legal when its turn comes.

Source site s . If $s = i$: $i \in S$ and no generator of J fills or empties i before a_i , so i is occupied. If $s = i+r$ with $r \geq 1$: when $\varepsilon_r = 1$, a_s precedes a_{s-1} and $s \in S$ by (4), so s is still occupied; when $\varepsilon_r = 0$, a_{s-1} precedes a_s and $s \notin S$, so a_{s-1} has already filled s .

Target site $s+1$. If $s = i+e-1$: $i+e \notin S$ and no generator of J fills $i+e$ before a_{i+e-1} , so it is empty. If $s = i+r$ with $r \leq e-2$: when $\varepsilon_{r+1} = 0$, a_s precedes a_{s+1} and $s+1 \notin S$, so $s+1$ is still empty; when $\varepsilon_{r+1} = 1$, a_{s+1} precedes a_s and $s+1 \in S$, but a_{s+1} has already emptied it.

So every firing is legal, for *any* compatible linear order. Each a_s removes a bead from s and puts one at $s+1$; summing over $s \in J$ the intermediate sites telescope and the net change of the bead set is $-i, +(i+e)$. The scalar collected is q for each $s \in J$ with $s = n-1$, i.e. $q^{[n-1 \in J]}$, and $|J| = e \leq n-1$ so this exponent is 0 or 1. Finally $|\varepsilon| = \#\{s : i+s \in S, 1 \leq s \leq e-1\}$ by (4). $\square$

Theorem 5 (The t -family lies in $\mathcal{A}_n[t]$). *Let $1 \leq e \leq n-1$ and set*

$$R_e(t) := \sum_{i \in \mathbb{Z}/n\mathbb{Z}} \sum_{\varepsilon \in \{0,1\}^{e-1}} t^{|\varepsilon|} a_{[i, i+e-1], \varepsilon} \in \mathcal{A}_n[t]. \quad (5)$$

Then $R_e(t)$ acts on $\bigoplus_k \mathcal{QH}^*(\text{Gr}_{kn})$ exactly as the cylindric height-graded e -ribbon adder $\mathcal{R}_e(t)$ of (3). Moreover, for fixed i the 2^{e-1} summands act on pairwise disjoint sets of basis vectors, so the grading by $|\varepsilon|$ is the grading by height, term by term.

Proof. Fix S and i with $i \in S$, $i + e \notin S$. The occupancy pattern of S on the interior sites $i + 1, \dots, i + e - 1$ determines a unique $\varepsilon \in \{0, 1\}^{e-1}$ via (4), and by Lemma 4 exactly that summand is nonzero on σ_S ; it contributes $t^{|\varepsilon|} q^{[n-1 \in J] \sigma_{(S \setminus \{i\}) \cup \{i+e\}}}$, which is the $j = i$ term of (3). If $i \notin S$ or $i + e \in S$ then every summand at that i vanishes, matching (3). Summing over i gives the claim, and disjointness of supports is (4). $\square$

Remark 6. Theorem 5 is compatible with, and sharpens, Postnikov’s own l.2806: “the map $\kappa \mapsto \mathbf{a}_\kappa$ gives a one-to-one correspondence between cylindric shapes (modulo the ‘ $\sim$ ’-equivalence) and non-vanishing monomials in the algebra $\mathcal{A}_n$ ”, where $\sim$ (ll.2795–2803) is generated by independent south-east shifts of connected components. Restricted to *connected* shapes on e consecutive diagonals — i.e. to e -ribbons — that correspondence is exactly our $\varepsilon \leftrightarrow a_{J,\varepsilon}$, because an e -ribbon is determined by its word of $e - 1$ steps in $\{\text{right}, \text{down}\}$. What is new is the identification of the *statistic*: the down-steps of the ribbon are the “up”-oriented edges of the heap, hence $\text{ht} = |\varepsilon|$, hence the entire t -line is an orientation generating function inside one algebra.

Corollary 7 (Postnikov’s $\mathbf{e}$ and $\mathbf{h}$ are the two ends of the t -line). *With $\mathbf{e}_r, \mathbf{h}_r$ as in [1, ll.2853–2857],*

$$R_e(0) = \sum_i \prod_{J=[i, i+e-1]}^\circ a = (\text{connected part of } \mathbf{h}_e), \quad [t^{e-1}]R_e(t) = \sum_i \prod_J^\circ a = (\text{connected part of } \mathbf{e}_e).$$

Equivalently: a cylindric horizontal e -strip is a disjoint union of non-adjacent height-0 ribbons and a cylindric vertical e -strip is a disjoint union of non-adjacent height-maximal ribbons, and $\mathbf{h}_e, \mathbf{e}_e$ are obtained from $R_\bullet(0), R_\bullet(\infty)$ by allowing several non-adjacent runs.

Proof. Immediate from (2) and Lemma 4: the runs of a proper subset I are non-adjacent, hence their monomials commute and the bead moves they perform are disjoint; $\varepsilon = \mathbf{0}$ gives height 0 on each run and $\varepsilon = \mathbf{1}$ gives height $\ell_a - 1$ on a run of length ℓ_a . $\square$

Corollary 8 (The duality $\mathbf{e} \leftrightarrow \mathbf{h}$ is $t \mapsto 1/t$). *Let ρ be the Grassmannian duality $P_{kn} \rightarrow P_{n-k,n}$, $\lambda \mapsto \lambda'$, which on beads is $\rho(S) = \{n - 1 - j \bmod n : j \notin S\}$. Then*

$$\rho \mathcal{R}_e(t) \rho^{-1} = t^{e-1} \mathcal{R}_e(1/t).$$

This is the cylindric form of the functional equation $\Omega R_e(t) \Omega = t^{e-1} R_e(1/t)$ (PROVE 2026-08-30). It is verified symbolically for $3 \leq n \leq 7$, all $1 \leq k \leq n - 1$ (§6, Check 5); a proof is immediate from Lemma 3 (ρ exchanges arm and leg) but is not written out here. Corollaries 7 and 8 together say: the $\mathbf{e}/\mathbf{h}$ duality of $\mathcal{A}_n$ is the involution $t \mapsto 1/t$ of the ribbon t -line, exchanging its two endpoints.

4 Theorem B: the parameter is $t = -1$

Corollary 7 places $\mathbf{h}$ and $\mathbf{e}$ at $t = 0$ and $t = \infty$. The interior of the t -line carries the structure that makes Postnikov’s Lemma an *identity* rather than a definition.

Theorem 9 (verified, ranges as stated). *Let $2 \leq k \leq n - 2$.*

(i) (Rigidity.) $t = -1$ is the unique value of t for which $[\mathcal{R}_e(t), \mathbf{h}_j] = 0$ for all $1 \leq e \leq n-1$ and all $1 \leq j \leq n-1$. Verified for $4 \leq n \leq 6$. (For $k = 1$ and $k = n-1$ the commutator vanishes identically in t ; the space is too small to see a constraint, and those cases carry no evidence either way.)

(ii) (Newton.) Writing $p_e := \mathcal{R}_e(-1)$, the operators $p_1, \dots, p_{n-1}$ commute with one another and with $\mathbf{h}_1, \dots, \mathbf{h}_{n-1}$, and

$$r \mathbf{h}_r = \sum_{e=1}^r p_e \mathbf{h}_{r-e} \quad (1 \leq r \leq n-1).$$

Verified for $3 \leq n \leq 7$ and all $1 \leq k \leq n-1$, exactly and symbolically in $\mathbb{Z}[q]$.

(iii) (The defect.) At $r = n$ the identity fails, with

$$\sum_{e=1}^{n-1} p_e \mathbf{h}_{n-e} = (-1)^{k-1} (n-k) q \cdot \text{Id} \quad \text{on } \text{QH}^*(\text{Gr}_{kn}).$$

Verified for $3 \leq n \leq 7$, all k .

Proposition 10 (Consequence: the quantum Murnaghan–Nakayama rule). *Assume Theorem 9(ii). Then for $1 \leq e \leq n-1$, $\mathcal{R}_e(-1)$ is quantum multiplication by p_e on $\text{QH}^*(\text{Gr}_{kn})$:*

$$\mathcal{R}_e(-1) \sigma_\lambda = p_e * \sigma_\lambda.$$

Proof. By Postnikov’s quantum Pieri corollary (`cor:pieri-AN`, ll.2935–2947) $\mathbf{h}_j$ acts as quantum multiplication by $h_j = \sigma_j$. Let $\varphi : \Lambda[q] \rightarrow \text{QH}^*(\text{Gr}_{kn})$ be the ring homomorphism determined by $e_i \mapsto \sigma_{1^i}$ (zero for $i > k$); this is well defined because $\Lambda = \mathbb{Z}[e_1, e_2, \dots]$ is free, and by the presentation `eq:QH` (ll.500–507) it is surjective and sends $h_j \mapsto \sigma_j$ for $1 \leq j \leq n-1$. Newton’s identity $r \mathbf{h}_r = \sum_{e \leq r} p_e \mathbf{h}_{r-e}$ holds in Λ , hence its φ -image holds in $\text{QH}^*(\text{Gr}_{kn})$. Now induct on e . For $e = 1$, Theorem 9(ii) at $r = 1$ gives $\mathcal{R}_1(-1) = \mathbf{h}_1 = M_{p_1}$. For $e = r \leq n-1$, subtracting the two Newton relations,

$$\mathcal{R}_r(-1) = r \mathbf{h}_r - \sum_{e=1}^{r-1} \mathcal{R}_e(-1) \mathbf{h}_{r-e} = M_{r \mathbf{h}_r} - \sum_{e=1}^{r-1} M_{p_e} M_{\mathbf{h}_{r-e}} = M_{r \mathbf{h}_r - \sum_{e < r} p_e \mathbf{h}_{r-e}} = M_{p_r}. \quad \square$$

Remark 11 (why $t = -1$ and not $t = 0$). The brief anticipated a tension: $\mathcal{A}_n$ is *nil* ($a_i^2 = 0$), which in most such families is a $t \rightarrow 0$ degeneration, so $t = -1$ and “nil” might be incompatible. Lemma 4 shows the tension is not there. The nil relations constrain the *support* of a monomial — which occupancy patterns (4) it can act on — and not its height; heights $0, \dots, e-1$ all occur, in the 2^{e-1} different commutation classes with the same support J . What is true is that each individual monomial is supported on a single height; the t -grading is recovered only after summing over the commutation classes, which is exactly (5).

5 Theorem C: where the truncation comes from

Postnikov’s Lemma (`lem:e-h-commute-in-AN`, ll.2882–2893) states that $\mathbf{e}_i, \mathbf{h}_j$ commute pairwise, that $\mathbf{e}_i \mathbf{h}_j = 0$ for $i + j > n$, and that

$$\left(1 + \sum_{i=1}^{n-1} \mathbf{e}_i t^i\right) \left(1 + \sum_{j=1}^{n-1} \mathbf{h}_j (-t)^j\right) = 1 + \left(\sum_{m=1}^{n-1} (-1)^{n-m} \mathbf{e}_m \mathbf{h}_{n-m}\right) t^n. \quad (6)$$

Theorem 12. On $\mathrm{QH}^*(\mathrm{Gr}_{kn})$, $1 \leq k \leq n-1$:

(i) the correction term of (6) is the scalar operator

$$\sum_{m=1}^{n-1} (-1)^{n-m} \mathbf{e}_m \mathbf{h}_{n-m} = (-1)^{n-k} q \cdot \mathrm{Id};$$

(ii) it is nothing but the single quantum relation $\mathbf{e}_k \mathbf{h}_{n-k} = q$, all other terms of the sum being zero;

(iii) on the ribbon side, the images under φ of the two “missing” degree- n objects are

$$\varphi(h_n) = (-1)^{k-1} q, \quad \varphi(p_n) = (-1)^{k-1} k q,$$

and $\varphi(p_n)$ is exactly the value of the full-wrap bead move: each of the k beads travels n steps round the cylinder, jumping over the other $k-1$ beads and crossing the seam once, giving $k \cdot t^{k-1} q|_{t=-1}$.

Proof. (i) and (ii). By Postnikov’s presentation $\mathbf{eq}:\mathrm{QH}(\mathrm{ll}.500\text{--}507)$, $\mathrm{QH}^*(\mathrm{Gr}_{kn}) = \mathbb{Z}[q, e_1, \dots, e_k, h_1, \dots, h_{n-k}] / \langle e(t)1 + (-1)^{n-k} q t^n \rangle$, with $e_i = 0$ for $i > k$ and $h_j = 0$ for $j > n-k$. The coefficient of t^n of the defining relation reads $\sum_{m=0}^n (-1)^{n-m} e_m h_{n-m} = (-1)^{n-k} q$. In the sum, $e_m = 0$ unless $m \leq k$ and $h_{n-m} = 0$ unless $n-m \leq n-k$, i.e. $m \geq k$; so the only surviving term is $m = k$, giving $(-1)^{n-k} e_k h_{n-k} = (-1)^{n-k} q$, i.e. $e_k h_{n-k} = q$. Since $1 \leq k \leq n-1$, the index $m = k$ lies in the range $1 \leq m \leq n-1$ of (6), and all other terms vanish. By `cor:pieri-AN` the operators $\mathbf{e}_m, \mathbf{h}_j$ are the corresponding quantum multiplications, so the operator identity follows.

(iii). With φ as in Proposition 10: in Λ we have $h(-t) = e(t)^{-1}$, whereas in $\mathrm{QH}^*(\mathrm{Gr}_{kn})$ $h_{\mathrm{QH}^*}(-t) = e(t)^{-1}(1 + (-1)^{n-k} q t^n)$. Hence, writing $u = -t$,

$$\varphi(h(u)) = h_{\mathrm{QH}^*}(u)(1 - (-1)^k q u^n + O(u^{2n})),$$

and taking the coefficient of u^n , using $h_{\mathrm{QH}^*,n} = 0$ and $h_{\mathrm{QH}^*,0} = 1$, gives $\varphi(h_n) = (-1)^{k-1} q$. Newton at $r = n$ in Λ , pushed through φ , then gives

$$\varphi(p_n) = n \varphi(h_n) - \sum_{e=1}^{n-1} \varphi(p_e) \varphi(h_{n-e}) = n(-1)^{k-1} q - (-1)^{k-1} (n-k) q = (-1)^{k-1} k q,$$

where the middle sum is evaluated by Proposition 10 together with Theorem 9(iii). The stated bead count gives the same number: for each of the k beads j , the move $j \mapsto j+n = j$ is legal (the site is vacated by the bead itself), it passes the other $k-1$ beads and crosses the seam once, contributing $t^{k-1} q$; at $t = -1$ the total is $k(-1)^{k-1} q$. $\square$

Corollary 13 (the answer to the truncation question). *The vanishing $\mathbf{e}_i \mathbf{h}_j = 0$ for $i+j > n$ and the correction term of (6) are **not** features of the t -family. They are features of the index set. Postnikov’s index set is $\{\text{proper subsets of } \mathbb{Z}/n\mathbb{Z}\}$; the ribbon family’s index set is $\{\text{cylindric ribbons}\}$, which contains the wrap-around ribbon of size n excluded by “proper”. On the semi-infinite wedge, whose index set is $\mathbb{Z}$, there is no wrap and hence no truncation; the mechanism that produces one is precisely the surjection $\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$.*

Remark 14 (one relation, four faces). The relation $e_k h_{n-k} = q$ — the only difference between $H^*(\mathrm{Gr}_{kn})$ and $\mathrm{QH}^*(\mathrm{Gr}_{kn})$ — appears in this paper in four guises: as the correction term $(-1)^{n-k} q$ of (6) (Theorem 12(i)); as the failure of Newton’s identity at $r = n$, of size $(-1)^{k-1} (n-k) q$ (Theorem 9(iii)); as $\varphi(h_n) = (-1)^{k-1} q$; and as $\varphi(p_n) = (-1)^{k-1} k q$, the full-wrap ribbon (Theorem 12(iii)). The arithmetic $n(-1)^{k-1} q = (-1)^{k-1} (n-k) q + (-1)^{k-1} k q$ that ties the last three together is the whole quantum content, read off the cylinder as “ n steps = $(n-k)$ gaps + k beads”.

6 Computational verification

All checks are exact and symbolic (SymPy over $\mathbb{Z}[q, t]$), in `proofs/code-q237/` (`anTL.py`, `ribbon_check.py`). The model implements $\mathcal{A}_n$ from the presentation (1) *via its action*, i.e. a_i is the bead move, and the relations (1) are consequences rather than inputs. Two independent routes to each object are compared.

1. **Abacus dictionary (Lemma 3).** Non-cyclic bead moves $j \rightarrow j + e$ versus the Young-diagram skew shape, checked for size, connectedness, absence of 2×2 , rows $-1 = h$ and cols $-1 = e - 1 - h$. All $n \leq 9$, all k , all S , all legal moves: **0 violations**.
2. **Theorem 5.** Orientation sum (5) versus the independent bead definition (3). $2 \leq n \leq 9$, all $0 \leq k \leq n$, all $1 \leq e \leq n - 1$: **0 mismatches**.
3. **Disjoint supports / nonvanishing.** For $e = 4$, $n \in \{6, 7\}$, $k \in \{2, 3, 4\}$: the $2^{e-1} = 8$ monomials per interval have pairwise disjoint state-supports (**0 overlaps**); those that vanish identically are exactly the ones violating $|\varepsilon| + 1 \leq k$ or $e - |\varepsilon| \leq n - k$, as (4) predicts.
4. **Postnikov's Lemma, reproduced.** For $3 \leq n \leq 7$, all k : the coefficients of $t^1, \dots, t^{n-1}$ in the left side of (6) vanish; $\mathbf{e}_i \mathbf{h}_j = 0$ for $i + j > n$; the t^n coefficient equals Postnikov's stated right-hand side; and that scalar is $(-1)^{n-k}q$.
5. **Corollary 8.** ρ with $\rho(S) = \{n - 1 - j : j \notin S\}$ intertwines $\mathcal{R}_e(t)$ on (n, k) with $t^{e-1}\mathcal{R}_e(1/t)$ on $(n, n - k)$, for $3 \leq n \leq 7$, all k , $e = \min(3, n - 1)$.
6. **Theorem 9.** Commutation $[\mathcal{R}_e(t), \mathbf{h}_j]$ solved for t entrywise: the common root set is $\{-1\}$ for $4 \leq n \leq 6$, $2 \leq k \leq n - 2$, and unconstrained for $k \in \{1, n - 1\}$. Newton verified for $3 \leq n \leq 7$, all k , all $r \leq n - 1$; defect at $r = n$ equals $(-1)^{k-1}(n - k)q$ in every case.
7. **Full-wrap value.** The $e = n$ bead move evaluated at $t = -1$ equals $(-1)^{k-1}kq$ for $3 \leq n \leq 7$, all k .

7 Gaps

These are stated as propositions, not as descriptions, so that each is falsifiable on its own.

- (G1) **Theorem 9 is verified, not proved.** Both (i) rigidity at $t = -1$ and (ii) Newton's identity in $\mathcal{A}_n$ are open. Proposition 10 derives the quantum Murnaghan–Nakayama rule *from* (ii); it does not prove (ii). Equivalently one may prove $[\mathcal{R}_e(-1), \mathbf{h}_j] = 0$ directly, since with quantum Giambelli (`eq:Giambelli`, ll.534–540) an operator commuting with all $\mathbf{h}_j$ is multiplication by its value on $\sigma_\emptyset$. I expect a sign-reversing involution on pairs (cylindric horizontal strip, signed ribbon); I did not find it today.
- (G2) **The $q = 0$ half of (G1) is not the hard half.** At $q = 0$, $\mathcal{R}_e(-1) = M_{p_e}$ on $H^*(\text{Gr}_{kn})$ follows from the classical statement $R_e(-1) = M_{p_e}$ on Λ (PROVE 2026-09-03 c2, Q75) because multiplication by p_e preserves the ideal spanned by Schur functions outside the box. All the content of (G1) is in the q^1 terms, i.e. in the wrapping ribbons. I have *not* checked that the $q = 0$ reduction argument is airtight; it is written here as a plan, not as a proof.

- (G3) **Rigidity is untested at the boundary.** For $k = 1$ and $k = n - 1$ the commutator $[\mathcal{R}_e(t), \mathbf{h}_j]$ vanishes for all t . I have *not* determined whether this is a rank degeneracy of small spaces or a genuine statement about $\text{Gr}_{1,n} = \mathbb{P}^{n-1}$; until that is decided, the phrase “ $t = -1$ is unique” must carry the hypothesis $2 \leq k \leq n - 2$.
- (G4) **The convention of Remark 1 is a reading, not a quotation.** Postnikov’s displayed `e2` (1.2862) and his combinatorial description (1.2877) are incompatible under right-to-left composition; I resolved this in favour of the combinatorial description, because that is what `cor:pieri-AN` and the rest of the section use. A reader holding a different build of the paper should re-check 1.2862 before relying on (2). All line numbers here refer to the e-print I hold; they are internal locators, not published theorem numbers.
- (G5) **Corollary 8 is verified and sketched, not proved.** The sketch (“ ρ exchanges arm and leg”) is one line and I believe it; it is not written out.
- (G6) **Nothing here touches the S_ℓ -symmetry itself.** Theorem 5 identifies the operators; it does not give the commutativity argument that Postnikov’s ll.1460–1464 sketch, nor does it bear on the M-convexity programme (WZZ Problem 5.2), which concerns supports rather than operators.

Novelty statement

The operator-level bridge between cylindric combinatorics and integrable systems is not new, and no sentence here may claim it. Korff–Stroppel [2] (1.272) explicitly “*reprov[e] a result of Postnikov*” by realising $\mathcal{A}_n$ on a fermionic Fock space — the generators a_i of this paper are, in their language, “one fermion hopping in clockwise direction on the circle” (1.2474). Korff–Vasilev, *Equivariant Quantum Cohomology of Grassmannians via the Clifford algebra* (`arXiv:2606.06352`, May 2026), go further: §`sec:cylloops` (1.1483) puts Postnikov’s cylindric loops in periodic Maya diagrams — the bead model of this paper — and App. B (1.1734) gives the equivariant analogue of $\mathcal{A}_n$ as $\varphi(u_i) = \psi_{i+1}^* \psi_i + y_i \psi_i^* \psi_i$. Benkart–Meinel, *The center of the affine nil-Temperley–Lieb algebra* (`arXiv:1505.02544`), study exactly this algebra through “a faithful representation of it on fermionic particle configurations”. An OpenAlex query for “affine nil-Temperley–Lieb” returns sixty works, *none* of which was in my sources index before this morning.

I therefore make no priority claim of any kind. What I claim is that the following three statements were derived in this session and are, to my knowledge, not written down in the sources I hold: the identification of the ribbon *height* with the orientation statistic of the heap (Theorem 5); the consequent realisation of the whole t -line $R_e(t)$ — not just its two endpoints — inside $\mathcal{A}_n[t]$; and the location of `eq:e-h-AN`’s truncation as the $e = n$ wrap-around ribbon (Theorem 12). I have **not** checked any of the three against 2606.06352, 1505.02544, or the sixty-work nil-Temperley–Lieb literature; that check is owed, and until it is done each of the three should be read as “derived here”, not “new”.

In particular the following sentences are forbidden in any draft leaving this container: “the operator-level cylindric–integrable bridge is new”; “the first combinatorial proof of Postnikov’s symmetry”; “Postnikov’s proof is unwritten” (the sketch at ll.1460–1464 is unwritten; the algebra, the lemma and the identity are written, at ll.2837–2893).

References

- [1] A. Postnikov, *Affine approach to quantum Schubert calculus*, `arXiv:math/0205165v2`. Cited by internal $\text{\LaTeX}$ labels and line numbers of the e-print source at `sources/math0205165-postnikov/postnikov-affine-v2.tex`, read at source on 2026-09-23.
- [2] C. Korff and C. Stroppel, *The $\widehat{\mathfrak{sl}}(n)_k$ -WZNW fusion ring: a combinatorial construction and a realisation as quotient of quantum cohomology*, *Adv. Math.* 225 (2010) 200–268, `arXiv:0909.2347`. Line numbers refer to `sources/0909.2347-korff-stroppel/src/revisedversion4.tex`.